

# Representative Structural Calculation Package

Apartment frame equivalent

| Item            | Value                                                        |
|-----------------|--------------------------------------------------------------|
| Project         | Apartment frame equivalent                                   |
| Purpose         | Representative elastic analysis and preliminary design trace |
| Generated       | 2026-06-27T00:00:00.000Z                                     |
| Package version | m42-calculation-package                                      |
| Source model    | 07-apartment-frame-equivalent                                |

|       |         |       |        |           |           |
|-------|---------|-------|--------|-----------|-----------|
| Nodes | Members | Loads | Combos | Max disp. | Max util. |
| 126   | 270     | 756   | 28     | 4.173 mm  | 0.358     |

## Model And Design Basis

| Item            | Value                                                                        |
|-----------------|------------------------------------------------------------------------------|
| Building type   | apartment                                                                    |
| Description     | Six-story elongated residential block represented as an equivalent 3D frame. |
| Occupancy       | Residential                                                                  |
| Bounds X/Y/Z    | 25.00 / 7.600 / 17.40                                                        |
| Analysis status | OK                                                                           |
| Design status   | OK                                                                           |
| Package audit   | OK                                                                           |

## Load Case Trace

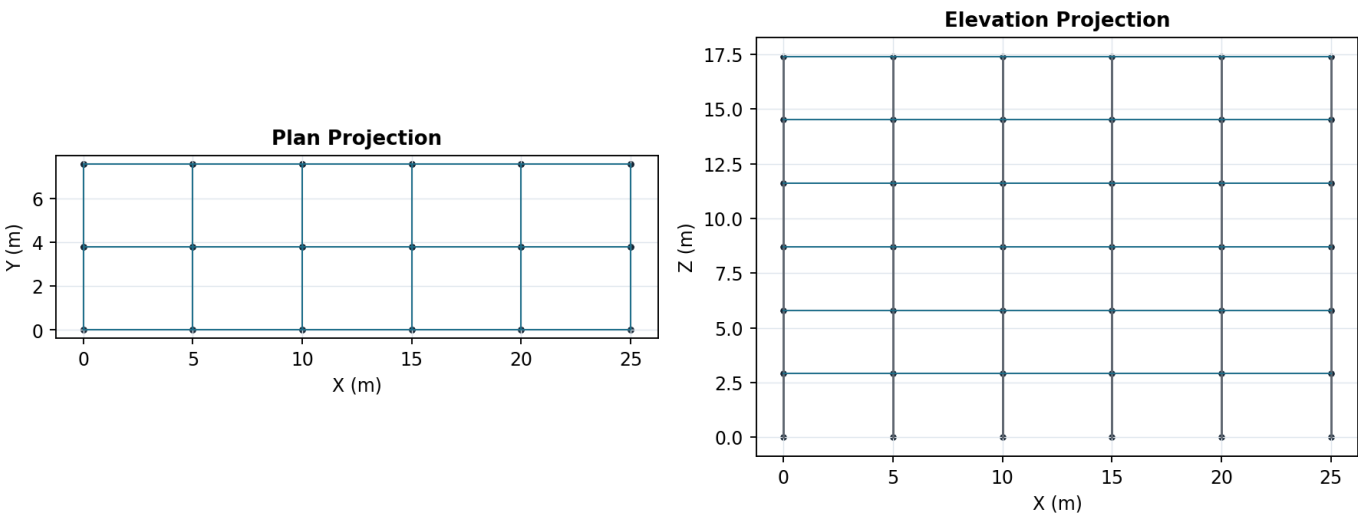
| Case | Type    | Loads | Fx        | Fy        | Fz       |
|------|---------|-------|-----------|-----------|----------|
| D    | dead    | 162   | 0.000 kN  | 0.000 kN  | -3420 kN |
| L    | live    | 162   | 0.000 kN  | 0.000 kN  | -1482 kN |
| WX   | wind    | 108   | 59.51 kN  | 0.000 kN  | 0.000 kN |
| WY   | wind    | 108   | 0.000 kN  | 195.75 kN | 0.000 kN |
| EX   | seismic | 108   | 344.94 kN | 0.000 kN  | 0.000 kN |
| EY   | seismic | 108   | 0.000 kN  | 344.94 kN | 0.000 kN |

## Load Derivation

| Story | Area   | D total   | L total   | Wind X   | Wind Y   |
|-------|--------|-----------|-----------|----------|----------|
| 1     | 190.00 | 570.00 kN | 266.00 kN | 9.918 kN | 32.62 kN |
| 2     | 190.00 | 570.00 kN | 266.00 kN | 9.918 kN | 32.62 kN |
| 3     | 190.00 | 570.00 kN | 266.00 kN | 9.918 kN | 32.62 kN |
| 4     | 190.00 | 570.00 kN | 266.00 kN | 9.918 kN | 32.63 kN |
| 5     | 190.00 | 570.00 kN | 266.00 kN | 9.918 kN | 32.63 kN |

|   |        |           |           |          |          |
|---|--------|-----------|-----------|----------|----------|
| 6 | 190.00 | 570.00 kN | 152.00 kN | 9.918 kN | 32.62 kN |
|---|--------|-----------|-----------|----------|----------|

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Combination Analysis Results

| Combo          | Status | Max disp. | Max util. | Residual |
|----------------|--------|-----------|-----------|----------|
| KDS-ST-01      | OK     | 0.763 mm  | 0.088     | 0.000    |
| KDS-ST-02      | OK     | 0.995 mm  | 0.119     | 0.000    |
| KDS-ST-04-WX-P | OK     | 1.017 mm  | 0.116     | 0.000    |
| KDS-ST-04-WX-N | OK     | 1.017 mm  | 0.116     | 0.000    |
| KDS-ST-04-WY-P | OK     | 2.025 mm  | 0.149     | 0.000    |
| KDS-ST-04-WY-N | OK     | 2.025 mm  | 0.149     | 0.000    |
| KDS-ST-05-EX-P | OK     | 3.928 mm  | 0.182     | 0.000    |
| KDS-ST-05-EX-N | OK     | 3.928 mm  | 0.182     | 0.000    |
| KDS-ST-05-EY-P | OK     | 4.173 mm  | 0.187     | 0.000    |
| KDS-ST-05-EY-N | OK     | 4.173 mm  | 0.187     | 0.000    |
| KDS-ST-06-WX-P | OK     | 0.721 mm  | 0.070     | 0.000    |
| KDS-ST-06-WX-N | OK     | 0.721 mm  | 0.070     | 0.000    |
| KDS-ST-06-WY-P | OK     | 1.895 mm  | 0.103     | 0.000    |
| KDS-ST-06-WY-N | OK     | 1.895 mm  | 0.103     | 0.000    |
| KDS-ST-07-EX-P | OK     | 3.858 mm  | 0.136     | 0.000    |
| KDS-ST-07-EX-N | OK     | 3.858 mm  | 0.136     | 0.000    |
| KDS-ST-07-EY-P | OK     | 4.118 mm  | 0.142     | 0.000    |

|                |    |          |       |       |
|----------------|----|----------|-------|-------|
| KDS-ST-07-EY-N | OK | 4.118 mm | 0.142 | 0.000 |
|----------------|----|----------|-------|-------|

## Governing Members

| Rank | Member | Status | Util. | Check            | Combo          |
|------|--------|--------|-------|------------------|----------------|
| 1    | M100   | OK     | 0.358 | steel-deflection | KDS-ST-05-EY-N |
| 2    | M99    | OK     | 0.358 | steel-deflection | KDS-ST-05-EY-P |
| 3    | M94    | OK     | 0.358 | steel-deflection | KDS-ST-05-EX-N |
| 4    | M106   | OK     | 0.358 | steel-deflection | KDS-ST-05-EX-N |
| 5    | M93    | OK     | 0.358 | steel-deflection | KDS-ST-05-EX-P |
| 6    | M105   | OK     | 0.358 | steel-deflection | KDS-ST-05-EX-P |
| 7    | M101   | OK     | 0.358 | steel-deflection | KDS-ST-05-EY-P |
| 8    | M98    | OK     | 0.358 | steel-deflection | KDS-ST-05-EY-N |
| 9    | M95    | OK     | 0.358 | steel-deflection | KDS-ST-05-EX-P |
| 10   | M107   | OK     | 0.358 | steel-deflection | KDS-ST-05-EX-P |
| 11   | M104   | OK     | 0.358 | steel-deflection | KDS-ST-05-EX-N |
| 12   | M92    | OK     | 0.358 | steel-deflection | KDS-ST-05-EX-N |
| 13   | M102   | OK     | 0.355 | steel-deflection | KDS-ST-05-EY-N |
| 14   | M97    | OK     | 0.355 | steel-deflection | KDS-ST-05-EY-P |
| 15   | M108   | OK     | 0.355 | steel-deflection | KDS-ST-05-EY-P |
| 16   | M96    | OK     | 0.355 | steel-deflection | KDS-ST-05-EY-N |
| 17   | M103   | OK     | 0.355 | steel-deflection | KDS-ST-05-EY-P |
| 18   | M91    | OK     | 0.355 | steel-deflection | KDS-ST-05-EY-N |
| 19   | M76    | OK     | 0.327 | steel-deflection | KDS-ST-05-EX-P |
| 20   | M88    | OK     | 0.327 | steel-deflection | KDS-ST-05-EX-P |

## Steel, Connection, Foundation Summary

| Scope        | Rows | Warnings/NG | Max ratio |
|--------------|------|-------------|-----------|
| Steel review | 270  | 0           | 0.358     |
| Connections  | 270  | 54          | 1.842     |
| Foundations  | 18   | 54          | 1.000     |

## Quality Audit

| Audit item                    | Status         |
|-------------------------------|----------------|
| Load derivation attached      | OK             |
| Combination results available | OK             |
| Member checks available       | OK             |
| RC schedule available         | Not applicable |
| Steel schedule available      | OK             |
| Foundation review available   | OK             |

## Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.
- Full KDS equation trace for every generated load combination and project-specific exception.

- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.